

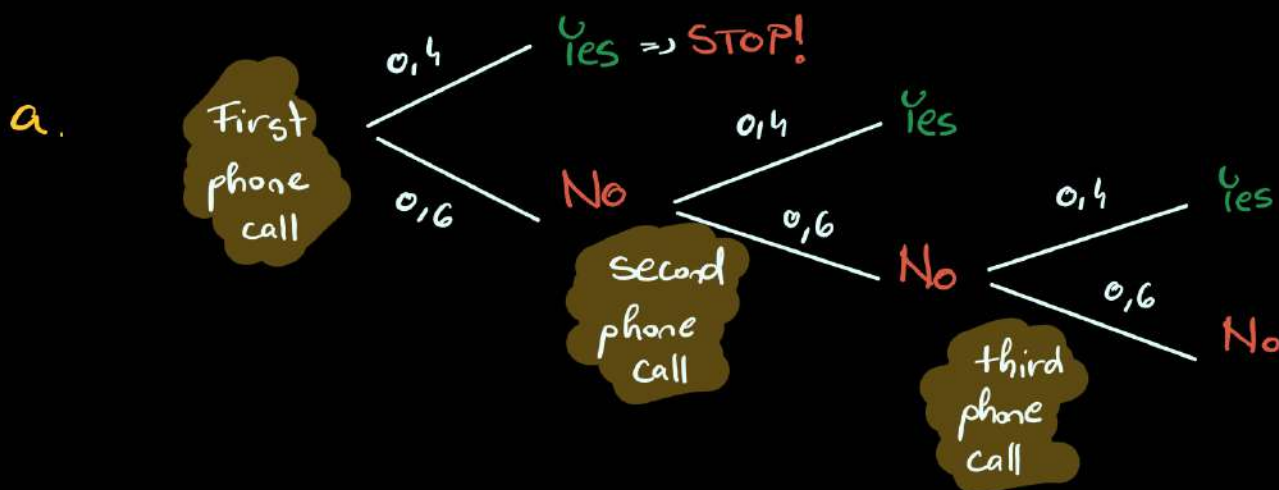
Discrete random variables

① For each phone call, the probability of obtaining information is 0,4. Let X - r.v., indicating the number of calls needed.

Find: a. PMF of X

b. CDF of X

c. The probability that at least 3 phone calls are needed



X - geometric r.v. with $p = 0,4$

$$P(X=1) = 0,4$$

$$P(X=2) = 0,4 \cdot 0,6$$

$$P(X=3) = 0,4 \cdot 0,6^2$$

$$\Rightarrow \underline{P_X(x)} = \begin{cases} 0,4 \cdot (0,6)^{x-1}, & x \in \{1, 2, 3, \dots\} \\ \text{zero} & , \text{ otherwise} \end{cases} =$$

$$= \begin{pmatrix} 1 & 2 & 3 & \dots & n & \dots \\ 0,4 & 0,4 \cdot 0,6 & 0,4 \cdot (0,6)^2 & \dots & 0,4 \cdot (0,6)^{n-1} & \dots \end{pmatrix}$$

$$b. F_X(x) = \begin{cases} \text{zero} & , x < 1 \\ 0,4 & , x \in [1,2) \\ 0,4 + 0,4 \cdot 0,6 & , x \in [2,3) \\ \dots & \dots \\ \dots & , x \in [n, n+1) \end{cases}$$

$$\begin{aligned} 0,4 + 0,4 \cdot 0,6 + 0,4 \cdot (0,6)^2 + \dots + 0,4 \cdot (0,6)^{n-1} &= \\ = 0,4 (1 + 0,6 + (0,6)^2 + \dots + (0,6)^{n-1}) &= \\ = 0,4 \cdot 1 \cdot \frac{(0,6)^n - 1}{0,6 - 1} = 1 - (0,6)^n \end{aligned}$$

$$c. \underline{P(X \geq 3)} = 1 - P(X < 3)$$

dacă avem mai

multe cazuri $\Rightarrow$ calculează $1 - \text{probabilitatea complementară}$

dacă nu, calculează probabilitatea

$$\underline{\cancel{P_X(0)} + P_X(1) + P_X(2)} = 0,4 + 0,4 \cdot 0,6 = 0,4 + 0,24 = 0,64$$

de interpretat contextul

problemei (putem începe

cu zero?) în acest caz NU

② 40% of admitted students will enroll

a. What's the probability that exact 6 students will enroll if the number of admissions increased with 10 units?

b. What's the probability that max 6 students will enroll if the number of admissions increased with 10 units?

c. if 40% of the admitted students will enroll, what's the probability that at least 12 from 15 students will enroll?

a. Because the first has to enroll and the second has to enroll
and the sixth has to enroll $\Rightarrow$ multiplication $\Rightarrow 40\% \cdot 60\%$

And because any six students from 10 have to enroll $\Rightarrow$

$\Rightarrow$ Combination

$$\Rightarrow \frac{\binom{10}{6} \cdot 0,4^6 \cdot 0,6^4}{C_{10}^6}$$

$$X \sim \text{Polynomial} : \left. \begin{array}{l} n=10 \\ h=6 \\ p=0,4 \end{array} \right| \Rightarrow P(X=6) = \frac{\binom{10}{6} \cdot 0,4^6 \cdot 0,6^4}{C_n^h \cdot p^h \cdot (1-p)^{n-h}}$$

$$\begin{aligned} \text{b. } P(X \leq 6) &= 1 - P(X > 6) = \\ &= 1 - (P(X=7) + P(X=8) + P(X=9) + P(X=10)) \\ &= 1 - \left(\binom{10}{7} \cdot 0,4^7 \cdot 0,6^3 - \binom{10}{8} \cdot 0,4^8 \cdot 0,6^2 - \right. \\ &\quad \left. \binom{10}{9} \cdot 0,4^9 \cdot 0,6^1 - \binom{10}{10} \cdot 0,4^{10} \cdot 0,6^0 \right) \end{aligned}$$

$$\text{c. } p=0,4 \quad n=15 \quad \Rightarrow X \sim \text{Binomial} \quad (p=0,4, n=15)$$

$$\begin{aligned}
 P(X \geq 12) &= \underline{P(X=12)} + \underline{P(X=13)} + \underline{P(X=14)} + \underline{P(X=15)} = \\
 &= \underline{\binom{15}{12} \cdot 0,4^{12} \cdot 0,3^3} + \underline{\binom{15}{13} \cdot 0,4^{13} \cdot 0,3^2} + \\
 &\quad \underline{\binom{15}{14} \cdot 0,4^{14} \cdot 0,3^1} + \underline{\binom{15}{15} \cdot 0,4^{15} \cdot 0,3^0}
 \end{aligned}$$

③ 25 % of members of Parliament will support the bill
5 member are contacted

a. What's the probability that at least 1 of the 5 members contacted will support the bill?

b. What's the probability that the majority of the 5 members contacted will support the bill?

a. $n=5$
 $h=5$
 $p=25\%$

ALWAYS - when we need to take a part of the set (at least one, exactly three) $\Rightarrow$ we have a Binomial R.V. !!

$\Rightarrow X \sim \text{Binomial}$

$$\begin{aligned}
 \Rightarrow P(X \geq 1) &= 1 - P(X < 1) = 1 - P(X=0) \\
 &= 1 - \binom{5}{0} \cdot 0,25^0 \cdot 0,75^5
 \end{aligned}$$

half or more
↓

$$\begin{aligned}
 \text{b. } P(X \geq \underline{3}) &= \underline{P(X=3)} + \underline{P(X=4)} + \underline{P(X=5)} = \\
 &= \underline{\binom{5}{3} \cdot 0,25^3 \cdot 0,75^2} + \underline{\binom{5}{4} \cdot 0,25^4 \cdot 0,75^1} + \underline{\binom{5}{5} \cdot 0,25^5 \cdot 0,75^0}
 \end{aligned}$$

④ The average number of requests received is 0,3/min.

a. What is the expected number of requests/hour?

b. What's the probability that more than one request will be received in 1 min?

c. What's the probability that no request will be received during 10 min interval?

a. When the question asks the number of something $\Rightarrow$
 $\Rightarrow$ Poisson r.v

$X \sim \text{Poisson}$

α - number of occurrence

T - time interval

$\Downarrow$

$$\lambda = \alpha \cdot T$$

$$P_X(x) = \begin{cases} \frac{e^{-\lambda} \cdot \lambda^x}{x!} & , x = 0, 1, 2, \dots \\ \text{zero} & , \text{otherwise} \end{cases}$$

0,3 request / min $\Rightarrow$ 0,3 · 60 requests / hour
we need in 1 hour

b. $T=1$
 $\alpha = 0,3 \quad | \Rightarrow \lambda = 0,3 \cdot 1 = 0,3$

$$P_X(x) = \begin{cases} \frac{e^{-0,3} \cdot 0,3^x}{x!} & , x = 0, 1, 2, \dots \\ \text{zero} & , \text{otherwise} \end{cases}$$

$$\underline{P(X \geq 1) = 1 - P(X=0) = 1 - \frac{e^{-0,3} \cdot 0,3^0}{0!} = 1 - e^{-0,3}}$$

$$c. \quad T=10 \quad \lambda=0,3 \quad \Rightarrow \quad \lambda = 10 \cdot 0,3 = 3$$

$$P_Y(x) = \begin{cases} \frac{e^{-3} \cdot 3^x}{x!}, & x = 0, 1, 2, \dots \\ \text{zero}, & \text{otherwise} \end{cases}$$

$$\underline{P(Y=0) = e^{-3}}$$

⑤ 86% of Italian households own a cell phone

a. What's the probability that from 20 families exactly 15 own a cell phone?

b. What's the probability that at least 18 families own a cell phone?

c. What's the probability that there are no more than 18 families to own a cell phone?

d. What's the probability that the number of families to own a cell phone is between 15 and 17?

e. If we ask 300 families, what is the number of families owning a cell phone? $E(X) = ?$ $\sigma(X) = ?$

$$a. \quad X \sim \text{binomial}, \quad n=20, \quad p=0.86$$

$$P(X=15)$$

$$b. \quad P(X \geq 18)$$

$$n=20 \rightarrow \text{for a, b, c, d}$$

c. $P(X \leq 18) = 1 - P(X > 18)$

d. $P(X=15) + P(X=16) + P(X=17)$

e. $E(X) \xrightarrow{\text{(for binomial)}} \underline{n \cdot p} \Rightarrow \text{here } n=300$

$\sigma(X) \xrightarrow{\text{(for binomial)}} \sqrt{n \cdot p \cdot (1-p)}$

After continuous r.v. $\Rightarrow$ First test at the seminary!!

⑥ R.v. X has CDF $\rightarrow F_X(x) = \begin{cases} 0 & , x < -1 \\ 0,2 & , x \in [-1, 0) \\ \textcircled{0,7} & , x \in [0, 1) \\ 1 & , x \geq 1 \end{cases}$

a. PMF of X

b. $B = \{ \quad \} \quad |x| > 0 \quad \rightarrow \text{without } x = \text{zero}$

$P_{X|B}(x) = ?$

c. $E(X|B) = ? \quad \text{var}(X|B) = ?$

$\rightarrow 0,2 + p \Rightarrow \underline{p=0,5}$

a. The intervals are given by the intervals between the events

$\Rightarrow X: \begin{pmatrix} -1 & 0 & 1 \\ 0,2 & \underset{0,5}{p} & 0,3 \end{pmatrix}$

PMF:

$P_X(x) = \begin{cases} 0,2 & , x = -1 \\ 0,5 & , x = 0 \\ 0,3 & , x = 1 \\ 0 & , \text{otherwise} \end{cases}$

$$b. P_{X|B}(x) = \begin{cases} \frac{P_X(x)}{P(B)} & , x \in B \\ \text{zero} & , \text{otherwise} \end{cases}$$

$$P(B) = P(X=-1) + P(X=1) = 0,2 + 0,3 = 0,5$$

$$P_{X|B}(x) = \begin{cases} \frac{0,2}{0,5} = \left(\frac{2}{5}\right) & , x = -1 \\ \frac{0,3}{0,5} = \left(\frac{3}{5}\right) & , x = 1 \\ \text{zero} & , \text{otherwise} \end{cases}$$

if we add them up $\Rightarrow$ the sum is 1 $\Rightarrow$ it is correct !!

$$c. \underline{E(X|B)} = \sum x_i \cdot P_{X|B}(x_i) = -1 \cdot \frac{2}{5} + 1 \cdot \frac{3}{5} = \underline{\frac{1}{5}}$$

$$\sigma(X|B) = \underline{E(X^2|B)} - \underline{[E(X|B)]^2} = \underline{\frac{2}{5} + \frac{3}{5} - \left(\frac{1}{5}\right)^2}$$

$$\quad \quad \quad (-1)^2 \cdot \frac{2}{5} + 1^2 \cdot \frac{3}{5} \quad \quad \quad = \left(\frac{1}{5}\right)^2$$